

TABLE II
HAND TOOLS

Name of tool	Function	Name of tool	Function
 Hand drill	It is needed for drilling holes of 1.5mm to 6.5mm diameter in cabinets/enclosures of electronics projects. The bit is attached to the drill by rotating the top (shiny portion in picture) anti-clockwise, placing the bit and then rotating it clockwise. You can also choose a motorised drill if cost is not a factor.	 IC chip extractor	An IC chip puller is used to pull chips off the PCBs without bending the pins of the IC.
 Drill bit sets	These bits are needed for drilling holes of 0.8mm to 6.5mm diameter in cabinets/enclosures of electronics projects. The bit can be attached to a hand drill or motorised drill. These come in sets. Fine 0.8mm and 1mm drill bits are required for drilling holes in PCBs.	 Illuminated magnifier	A magnifier like the one shown in the image that can magnify as well as illuminate the object under inspection is a must-have for inspecting solder joints for proper soldering, reading component markings and while dealing with surface mount device components.
 Needle files	Needle files of various types like square, semi-circle, triangular, round and flat are extremely useful for giving desired shapes or for imparting the perfect finish to the crudely-cut areas of electronic enclosures.	 15.2cm (6-inch) hand file set	This usually comes in sets that include various types like square, round, flat, triangular, semicircular, etc. It is useful in giving finishes to the otherwise crudely-cut electronic enclosures.
 Ten-piece trimming tool set	These form the essentials when it comes to the setting of trimmers, presets and the like, where the use of normal screwdrivers can cause hand capacitance effect.	 Lineman's plier	It has a large number of applications like cutting, bending and stripping wires. It is also useful in holding the bolt while tightening the nut.
 Fine screwdriver	This type of fine screwdrivers can be used for applying flux paste to PCBs.	 Needle-nose plier	It is useful in bending component leads prior to soldering these on PCBs. It can also be used to hold precision-fine bolts while tightening the nut.
 Solder-assist tool set	It is needed for cutting tracks of the PCB, holding SMD components while soldering, removing solder from the PCB, removing components after de-soldering, transforming holes to a larger desired size and aligning component leads.	 Nipper	Nippers are useful in cutting component leads in hard-to-reach areas where flush-side cutters cannot fit. Look for a nipper with one side of the cutting blade completely flat so that it cuts the leftover component lead with extreme precision.
 Hexagonal head L key set	These come in sets as well as loose and are very useful for opening grub screws.	 Open-end spanner set	Open-end spanners are useful for opening bolts from cabinets of large electronic appliances.
 Automatic wire stripper	These are useful for stripping off the insulation from single- or multi-stranded copper wires of 0.5mm to 6mm thickness.	 Large heavy-duty cutter	A large heavy-duty cutter can be used to cut PCBs, plastic cabinets and more.
 Baby vice	It is useful for holding PCBs or electronic enclosures while performing shaping, cutting, filing or drilling work on these.	 Helping third hand with magnifier	A helping third hand is used to hold the wires while tinning or joining these. It can also be used to hold small PCBs. It basically has two alligator clips that hold stuff together. It can be felt as an extension to our two hands, hence the name.
 Flush-side cutter	It is used to cut the leads of components after soldering work is done.	 Archimedes push drill	It is used to drill precision-fine holes of 0.8mm, 1mm or 1.2mm sizes on PCBs.
 Hacksaw	This tool is used to cut enclosures, PCBs and so on.	 15.2cm (6-inch) adjustable spanner	An adjustable spanner comes in handy to remove larger bolts from large cabinets of electronic appliances.
 Hammer with flat and point chisels	Hammer and flat chisels can be used to cut metal enclosures in a straight manner, while point chisels can be used to make a pre-mark before drilling.	 5mm flat screwdriver	It is useful in bending component leads after inserting components into the PCBs.
Hot glue gun	A hot glue gun is used in fixing cabinet-mounted parts to enclosures like switches, LEDs and so on.		